Combinatorics of Saddle Points in Deep Linear Networks

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Abstract

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1 Introduction: Informal statement of the main theorem

Flow of the paper: The main theorem will be the stratification of the critical loci via the stratification of the cyclic variety and its intersection wit the specialization. We introduce notation, we state the main theorem (without any quiver representation theory simply in terms of the rank patterns).

2 Related works

3 Setup and formal statement

The critical loci of rank r critical points in the loss landscape of a deep linear network admits a stratification parametrized by the rank of the critical points, rank patterns and a subset S of singular vectors of the teacher matrix M.

Definition 3.1 (Determinantal, cyclic, and critical loci)**.** Define

$$\Sigma_d^r := \{\theta \in \Omega : \operatorname{rank}(\mu(\theta)) = r\} \quad \textit{Determinantal variety},$$

$$\Gamma_d := \{(\theta, W_0) \in \Omega \times \mathrm{Mat}_{d_0 \times d_L} \mid W_{<\ell} W_0 W_{>\ell} = 0 \text{ for all } 1 \leq \ell \leq L\}, \quad \textit{Cyclic variety}$$

Let $\iota_S : \Omega \longrightarrow \Omega \times \mathrm{Mat}_{d_0 \times d_L}$ be an inclusion map such that $\iota_S(\theta) := (\theta, \Sigma_{XY} P_Q)$ and define:

$$\Gamma_d^r := \Gamma_d \cap (\Sigma_d^r \times \mathrm{Mat}_{d_0 \times d_L}), \qquad \Gamma_d^S := \iota_S^{-1}(\Gamma_d^r)$$

Let S be a subset of left-singular vectors of M, define the S-critical loci:

$$\mathcal{C}_d^S = \{\theta \in \Omega | \ \nabla_\theta L(\theta) = 0; \mu(\theta) = P_S W^\star\}$$

Definition 3.2 (Symmetry group)**.** For dimension vector $d = (d_0, ..., d_L)$ define the symmetry group:

$$G_d = \prod_{i=0}^{L} GL_{d_i}$$

Consider the action of G_d:

$$g.(\theta, W_0) = (g_L W_L g_{L-1}^{-1}, \ldots, g_1 W_1 g_0^{-1}, g_0 W_0 g_L^{-1})$$

and note that we have a well defined action of G_d over the cyclic variety from G_d-invariance of the equations defining the cyclic variety. The variety Γ_d^r also inherits an action of G_d from the action of G_d over $\Sigma_d^r \times \mathrm{Mat}_{d_0 \times d_L}$, since the rank is invariant under action of GL_d. A consequence of the action is that we can stratify the variety Γ_d^r into orbits (strata) O_ρ. In this paper we will show that the relevant geometric invariant ρ parametrizing the orbits are the cyclic rank patterns.

Definition 3.3. For $\phi = (\theta, W_0) \in \Gamma_d$ we define its *cyclic rank pattern* to be the array

$$\rho(\phi) = (r_{ij}(\phi))_{0 \leq i,j \leq L} \quad \text{where} \quad \rho_{ij}(\phi) := \operatorname{rank}\big(W_j ... W_i\big) \quad \text{for} \quad \text{and} \quad \rho_{ii}(\phi) = d_i \text{and} \quad \rho_{i+1i}(\phi) = \operatorname{rank}\big(W_i\big)$$

Theorem 3.4. Define the strata $\mathcal{O}_\rho^S := \mathcal{O}_\rho \cap \Gamma_{d,S}^r$ where $\mathcal{O}_\rho$ is a G_d-orbit of the cyclic variety Γ_d^r. Assuming that the data is non degenerate and the input covariance matrix is invertible then the loci of critical points of rank $r < r_{\max}$ admits the following stratification parameterized by subset S of singular vectors of the teacher with r elements and admissible cyclic rank patterns $\rho \in R_d^{\mathrm{orb}}$:

$$\mathcal{C}_d^r = \bigsqcup_{|S|=r} \mathcal{C}_d^S = \bigsqcup_{|S|=r} \bigsqcup_{\substack{\rho \in R_d^{\mathrm{orb}} \\ \operatorname{rank}(\rho) = \operatorname{rank}(\Sigma_{XY} P_Q)}} \mathcal{O}_\rho^S$$

4 Geometry: Rank pattern stratification

The purpose of this section is to relate the S-critical locus to the specialization of the cyclic variety obtained by fixing the closing arrow to be $\Sigma_{XY}P_Q$. Recall that

$$\Gamma_d^S = \iota_S^{-1}(\Gamma_d^r), \qquad \iota_S(\theta) = (\theta, \Sigma_{XY}P_Q).$$

Thus $\theta \in \Gamma_d^S$ precisely when

$$(\theta, \Sigma_{XY}P_Q) \in \Gamma_d^r.$$

Proposition 4.1 (The S-critical locus as a product fiber)**.** Assume that Σ_X is invertible, that

$$W^* = \Sigma_{YX}\Sigma_X^{-1},$$

and that $\operatorname{rank}(P_S W^*) = r$. Then, as subsets of Ω,

$$\mathcal{C}_d^S = \left\{\theta \in \Gamma_d^S \;\middle|\; \mu(\theta) = P_S W^*\right\}.$$

Equivalently,

$$\mathcal{C}_d^S = \Gamma_d^S \cap \mu^{-1}(P_S W^*) = \iota_S^{-1}(\Gamma_d^r) \cap \mu^{-1}(P_S W^*).$$

Proof. The point is that the gradient equations with $\mu(\theta) = P_S W^*$ are exactly the cyclic equations with closing arrow $\Sigma_{XY}P_Q$. Indeed,

$$\Sigma_X (P_S W^*)^\top - \Sigma_{XY} = -\Sigma_{XY}P_Q.$$

Therefore

$$\left(\nabla_{W_\ell}\mathcal{L}(\theta)\right)^\top = -2W_{<\ell}\Sigma_{XY}P_Q W_{>\ell}.$$

Thus the partial gradients vanish for all ℓ if and only if

$$W_{<\ell}\Sigma_{XY}P_Q W_{>\ell} = 0 \qquad \text{for all } 1 \le \ell \le L.$$

Together with $\mu(\theta) = P_S W^*$ and $\operatorname{rank}(P_S W^*) = r$, this is exactly the condition that $\theta \in \Gamma_d^S \cap \mu^{-1}(P_S W^*)$. See Appendix 8.1 for the full verification. □

Now assume that the rank r cyclic variety admits a finite cyclic rank-pattern decomposition

$$\Gamma_d^r = \bigsqcup_{\rho \in R} O_\rho^r,$$

where each O_ρ^r is locally closed in Γ_d^r. We assume that the rank-pattern pieces satisfy the frontier inclusion

$$\overline{O_\rho^r}^{\Gamma_d^r} \subseteq \bigcup_{\rho' \le \rho} O_{\rho'}^r,$$

where $\rho' \le \rho$ means that

$$\rho'_{ij} \le \rho_{ij} \qquad \text{for all } i, j.$$

This assumption is implied, for example, by the stronger equality

$$\overline{O_\rho^r}^{\Gamma_d^r} = \bigcup_{\rho' \le \rho} O_{\rho'}^r.$$

The decomposition of Γ_d^r induces a decomposition of the S-critical locus by pulling back along ι_S and then imposing the product constraint.

Proposition 4.2 (Induced cyclic rank-pattern decomposition of the critical locus)**.** Define

$$\mathcal{C}_\rho^S := \iota_S^{-1}(O_\rho^r) \cap \mu^{-1}(P_S W^*).$$

Then, after discarding empty pieces,

$$\mathcal{C}_d^S = \bigsqcup_{\rho \in R} \mathcal{C}_\rho^S.$$

Each $\mathcal{C}_\rho^S$ is locally closed in $\mathcal{C}_d^S$. Moreover, the relative closures satisfy

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} \subseteq \bigcup_{\rho' \leq \rho} \mathcal{C}_{\rho'}^S.$$

Proof. By Proposition 4.1,

$$\mathcal{C}_d^S = \iota_S^{-1}(\Gamma_d^r) \cap \mu^{-1}(P_S W^*).$$

Since the O_ρ^r form a disjoint decomposition of Γ_d^r, their pullbacks along ι_S, intersected with $\mu^{-1}(P_S W^*)$, form a disjoint decomposition of $\mathcal{C}_d^S$. Local closedness follows because preimages of locally closed sets under continuous maps are locally closed, and $\mu^{-1}(P_S W^*)$ is closed.

For the frontier statement, use the elementary topological inclusion

$$\overline{f^{-1}(A)} \subseteq f^{-1}(\overline{A})$$

for continuous f, applied to the restriction of ι_S to $\mathcal{C}_d^S$. Together with the frontier inclusion for O_ρ^r, this gives

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} \subseteq \bigcup_{\rho' \leq \rho} \mathcal{C}_{\rho'}^S.$$

The full argument is written in Appendix 8.2. □

Thus the S-critical locus inherits a coarse decomposition by cyclic rank patterns. The closure statement is deliberately one-sided: along degenerations inside $\mathcal{C}_d^S$, cyclic ranks can only decrease. We do not claim that every lower cyclic rank pattern appearing in $\mathcal{C}_d^S$ is reached as a degeneration inside the fixed-S critical locus. We see that we can stratify the critical locus by stratifying the cyclic variety. In the following sections we will show how to obtain such stratification using quiver representation theory.

5 Stratification of the cyclic variety with multiplicity patterns

Goal: Stratify the cyclic variety with cyclic rank patterns.

5.1 Essentials of Quiver representation theory

A quiver is a graph (I, E) with vertices I, edges E and an orientation. A quiver representation assigns a vector space to $V_i \cong \mathbb{R}^{d_i}$ and a linear map $\phi : V_i \to V_{i+1}$ with vector dimension d_i.

5.2 The cyclic Nakayama algebra

Throughout this subsection, put $n = L+1$, and take all indices modulo n. Unless otherwise stated, all A-modules are finite-dimensional left A-modules.

Let $Q = C_{L+1}$ be the cyclic quiver with vertex set

$$Q_0 = \{0, \dots, L\}$$

and arrows

$$a_l : l-1 \longrightarrow l, \qquad l \in \{0, \dots, L\},$$

where $a_0 : L \to 0$. We read products of arrows from right to left. For $1 \le l \le L$, define the length L path

$$p_l := a_{l-1} \cdots a_1 a_0 a_L \cdots a_{l+1}.$$

Thus p_l is the unique path of length L starting at l, ending at $l-1$, and passing through the closing arrow a_0.

Let

$$I = \langle p_1, \dots, p_L \rangle$$

and define the bound path algebra

$$A := kQ/I.$$

Equivalently, an A-module with dimension vector $d = (d_0, \dots, d_L)$ is a representation of Q on vector spaces $V_i \simeq k^{d_i}$, with linear maps

$$\phi_l : V_{l-1} \longrightarrow V_l,$$

satisfying the relations

$$\phi_{l-1} \circ \cdots \circ \phi_1 \circ \phi_0 \circ \phi_L \circ \cdots \circ \phi_{l+1} = 0, \qquad 1 \le l \le L.$$

We write $\mathrm{Rep}_d(A)$ for the corresponding representation variety.

Proposition 5.1. For $k = \mathbb{R}$, the cyclic variety Γ_d is identified with $\mathrm{Rep}_d(A)$ by

$$W_l = \phi_l, \qquad V_i \simeq \mathbb{R}^{d_i}.$$

Proof. Under this identification, the defining equations of Γ_d are exactly the relations generated by the paths $p_1, \dots, p_L$. Hence $\Gamma_d \cong \mathrm{Rep}_d(A)$. □

Let $R \subset kQ$ be the arrow ideal, i.e. the ideal generated by all arrows. Every path of length $L+1$ contains one of the paths $p_1, \dots, p_L$ as a subpath. Therefore

$$R^{L+1} \subset I.$$

It follows that A is finite-dimensional. Moreover,

$$J(A) = R/I,$$

so $J(A)$ is spanned by the nonzero paths of positive length, and

$$A/J(A) \cong \bigoplus_{i=0}^{L} k e_i.$$

In particular, the simple A-modules are

$$S_i := P_i / J P_i, \qquad P_i := A e_i.$$

Definition 5.2. A finite-length module U is called *uniserial* if its submodules are totally ordered by inclusion. Equivalently, U has a unique composition series.

A finite-dimensional algebra is called a Nakayama algebra if its indecomposable projective and indecomposable injective modules are uniserial.

Proposition 5.3. The algebra $A = kQ/I$ is a finite-dimensional Nakayama algebra. Its Kupisch series is

$$(c_0, \dots, c_L) = (L+1, L, \dots, L).$$

Proof. We first describe the indecomposable projectives. With our convention for path multiplication, $P_i = Ae_i$ is spanned by the nonzero paths starting at i. Let

$$q_{i,0} := e_i, \qquad q_{i,\ell} := a_{i+\ell} \cdots a_{i+1}$$

be the unique path of length ℓ starting at i. The path $q_{i,\ell}$ is nonzero precisely for

$$0 \le r < c_i,$$

where

$$c_0 = L+1, \qquad c_i = L \quad \text{for } 1 \le i \le L.$$

Indeed, the path starting at 0 of length L does not pass through a_0, so it is nonzero, whereas the path starting at 0 of length $L+1$ vanishes. If $i \neq 0$, then the path of length L starting at i is exactly the relation p_i, so it vanishes.

Thus

$$P_i = \langle q_{i,0}, q_{i,1}, \dots, q_{i,c_i-1} \rangle_k,$$

and

$$J^s P_i = \langle q_{i,s}, q_{i,s+1}, \dots, q_{i,c_i-1} \rangle_k, \qquad 0 \le s \le c_i.$$

In particular,

$$P_i \supset JP_i \supset J^2 P_i \supset \cdots \supset J^{c_i} P_i = 0.$$

Moreover,

$$J^s P_i / J^{s+1} P_i \cong S_{i+s}.$$

Hence this chain is a composition series.

It remains to check that it is the unique composition series. Let $N \subset P_i$ be a nonzero submodule. Since the idempotents e_v act as projections and each space $e_v P_i$ is at most one-dimensional, N is spanned by some subset of the basis elements $q_{i,\ell}$. If $q_{i,r} \in N$, then stability under the action of the arrows implies

$$q_{i,\ell+1}, q_{i,\ell+2}, \dots, q_{i,c_i-1} \in N.$$

Therefore

$$N = J^s P_i$$

for some s. Hence the submodules of P_i are totally ordered, and P_i is uniserial.

The modules $P_i = Ae_i$ form a complete set of indecomposable projective A-modules. The same argument applied to the opposite algebra shows that the indecomposable injective A-modules are also uniserial. Therefore A is a finite-dimensional Nakayama algebra. □

Lemma 5.4. Let B be a finite-dimensional Nakayama algebra. Then every indecomposable finite-dimensional B-module is uniserial.

Proof. This is the standard Artinian characterization of Nakayama algebras: for a finite-dimensional algebra B, the following are equivalent:

1. B is Nakayama;
2. every indecomposable finite-dimensional B-module is uniserial;
3. every finite-dimensional B-module is a direct sum of uniserial modules.

Since B is finite-dimensional, it is Artinian, so this characterization applies. □

Definition 5.5. For $i, j \in Q_0$, let

$$\delta(i,j) \in \{0, \dots, L\}$$

be the unique integer such that

$$j \equiv i + \delta(i,j) \pmod{L+1}.$$

Whenever

$$\delta(i,j) + 1 \le c_i,$$

define the interval module $M(i,j)$ by

$$M(i,j)_v = \begin{cases} k, & \text{if } v \equiv i + \ell \pmod{L+1} \text{ for all } 0 \le \ell \le \delta(i,j), \\ 0, & \text{otherwise,} \end{cases}$$

and

$$M(i,j)_{a_m} = \begin{cases} \mathrm{id}_k, & \text{if } m \equiv i + \ell \pmod{L+1} \text{ for all } 1 \le \ell \le \delta(i,j), \\ 0, & \text{otherwise.} \end{cases}$$

The condition $\delta(i,j) + 1 \le c_i$ is exactly the condition that the relations in I are satisfied.

Theorem 5.6. Every nonzero indecomposable finite-dimensional A-module is isomorphic to a unique interval module $M(i,j)$ with

$$\delta(i,j) + 1 \le c_i.$$

Equivalently, the indecomposable A-modules are given by:

$$M(i, i+\ell-1), \qquad 0 \le i \le L, \quad 1 \le \ell \le c_i,$$

where indices are taken modulo $L+1$.

Proof. Let M be a nonzero indecomposable finite-dimensional A-module. Since A is a finite-dimensional Nakayama algebra, the previous lemma implies that M is uniserial. Hence M has a unique maximal submodule, and therefore its top is simple:

$$\operatorname{top} M := M/JM \cong S_i$$

for a unique vertex i.

Let

$$\pi : P_i \twoheadrightarrow M$$

be the projective cover of M. Such a projective cover exists because A is finite-dimensional, and it must be P_i because $\operatorname{top} P_i \cong S_i$. Let $K = \ker \pi$. Then

$$M \cong P_i/K.$$

Since P_i is uniserial, its submodules are precisely

$$P_i,\ JP_i,\ J^2P_i, \dots,\ J^{c_i}P_i = 0.$$

Because $M \neq 0$, the kernel is not all of P_i. Thus there is a unique integer ℓ with $1 \leq \ell \leq c_i$ such that

$$K = J^\ell P_i.$$

Therefore

$$M \cong P_i/J^\ell P_i.$$

As a representation of the quiver, $P_i/J^l P_i$ has one-dimensional vector spaces at the vertices

$$i,\ i+1,\ \dots,\ i+\ell-1$$

and zero vector spaces elsewhere. The arrow maps along this interval are id_k, and all other arrow maps are zero. Hence

$$P_i/J^\ell P_i \cong M(i, i+\ell-1).$$

Conversely, for every i and every $1 \leq r \leq c_i$, the interval module $M(i, i+\ell-1)$ is isomorphic to $P_i/J^\ell P_i$. Since it is a nonzero quotient of a uniserial module, it is uniserial, hence indecomposable.

Finally, uniqueness follows because the top of $M(i, i+\ell-1)$ is S_i, which determines i, and its composition length is ℓ, which determines the endpoint $i+\ell-1$. This proves the classification. □

Definition 5.7. Let

$$c_0 = L+1, \qquad c_i = L \quad \text{for } 1 \leq i \leq L.$$

Define the set of admissible intervals by

$$\mathcal{I} := \{(i,\ell) \mid i \in \{0, \dots, L\},\ 1 \leq \ell \leq c_i\}.$$

For $(i,\ell) \in \mathcal{I}$, write

$$j(i,\ell) := i+\ell-1 \pmod{L+1}.$$

The corresponding interval module is

$$M_\ell(i) := M(i, j(i,\ell)).$$

Equivalently, $M_\ell(i)$ is supported on the nonzero path

$$i \to i+1 \to \cdots \to i+\ell-1,$$

with one-dimensional vector spaces along this path and identity maps along the arrows of the path.

The number of admissible intervals is

$$|\mathcal{I}| = c_0 + \sum_{i=1}^{L} c_i = (L+1) + L^2 = L^2 + L + 1.$$

Corollary 5.8 (Krull–Schmidt decomposition for $\mathrm{Rep}_d(A)$)**.** Every finite-dimensional A-module M admits a decomposition

$$M \cong \bigoplus_{(i,\ell)\in\mathcal{I}} M_\ell(i)^{\oplus m_{i,\ell}},$$

where $m_{i,\ell} \in \mathbb{Z}_{\geq 0}$, and only finitely many multiplicities are nonzero.

Moreover, the multiplicities $m_{i,\ell}$ are uniquely determined by M. Equivalently, the decomposition is unique up to permutation and isomorphism of the indecomposable summands.

Proof. Since A is finite-dimensional, the category of finite-dimensional A-modules is a Krull–Schmidt category. Hence every finite-dimensional A-module decomposes as a finite direct sum of indecomposable A-modules, and this decomposition is unique up to permutation and isomorphism of the indecomposable summands.

By the classification theorem, every nonzero indecomposable A-module is isomorphic to an interval module

$$M_\ell(i) \cong P_i/J^\ell P_i$$

for a unique pair

$$(i,\ell) \in \mathcal{I}.$$

Therefore every finite-dimensional A-module has a decomposition of the form

$$M \cong \bigoplus_{(i,\ell)\in\mathcal{I}} M_\ell(i)^{\oplus m_{i,\ell}}.$$

The uniqueness statement in Krull–Schmidt implies that the multiplicities $m_{i,\ell}$ are uniquely determined. □

5.3 Stratification with multiplicity patterns

Definition 5.9 (Multiplicity Patterns)**.** Consider a A-module M of vector dimension d and its Krull-Schmidt decomposition into $M_\ell(i)$ for $(i,\ell) \in \mathcal{I}$. For each vertex k the dimension d_k of the representation M at vertex k satisfies the dimension constraint:

$$d_k \;=\; \sum_{(i,\ell)\in\mathcal{I}} m_{i,\ell}\,\mathbf{1}\{k \in \{i, i+1, \ldots, i+\ell\} \pmod{L+1}\} \quad k = 0, \ldots, L$$

Define the set of *multiplicity patterns* as

$$\mathcal{M}_d^+ := \left\{ (m_{i,\ell}) \in \mathbb{Z}_{\geq 0}^{\mathcal{I}} \;\middle|\; d_k \;=\; \sum_{(i,\ell)\in\mathcal{I}} m_{i,\ell}\,\mathbf{1}\{k \in \{i, i+1, \ldots, i+\ell\} \pmod{L+1}\} \quad k = 0, \ldots, L \right\}.$$

We have a functorial equivalence between the isomorphism classes of A-modules and the G_d orbits of Γ_d. Given that all A-modules have a unique Krull-Schmidt decomposition indexed by the $m(i,j)$ then all G_d-orbits are indexed by the multiplicity patterns. In other words we have the following stratification:

Proposition 5.10 (Orbit stratification of $\Gamma_{d,L+1}$ by indecomposable Nakayama modules)**.** Fix $L \geq 1$ and let $Q = C_{L+1}$ be the oriented cyclic quiver on vertices $\{0, \ldots, L\}$. Let $A := kQ/I$ be the Nakayama quiver algebra, where I is the ideal generated by the length-L. The variety Γ_d decomposes as a disjoint union of these orbits:

$$\Gamma_d \;=\; \bigsqcup_{m\in\mathcal{M}_d^+} \mathcal{O}_m \;=\; \bigsqcup_{m\in\mathcal{M}_d^+} G_d \cdot \phi_m,$$

where ϕ_m is any representative whose associated A-module has multiplicity pattern m. In particular, the G_d-orbits in Γ_d are parameterized by the multiplicity data of indecomposable A-modules, i.e. by the set $\mathcal{M}_d^+$.

5.4 From multiplicity patterns to rank patterns

5.5 Specialization to fixed closing arrow

6 Stratification of the critical loci

7 Discussion

8 Appendix

8.1 Detailed proof of Proposition 4.1

We prove the equality

$$\mathcal{C}_d^S = \Gamma_d^S \cap \mu^{-1}(P_S W^*) = \iota_S^{-1}(\Gamma_d^r) \cap \mu^{-1}(P_S W^*)$$

as an equality of subsets of Ω.

Recall that

$$\iota_S(\theta) = (\theta, \Sigma_{XY} P_Q),$$

and therefore

$$\Gamma_d^S = \iota_S^{-1}(\Gamma_d^r).$$

Thus $\theta \in \Gamma_d^S$ means that

$$(\theta, \Sigma_{XY} P_Q) \in \Gamma_d^r.$$

Equivalently,

$$\operatorname{rank}(\mu(\theta)) = r$$

and

$$W_{<\ell} \Sigma_{XY} P_Q W_{>\ell} = 0 \qquad \text{for all } 1 \le \ell \le L.$$

We first prove

$$\mathcal{C}_d^S \subseteq \Gamma_d^S \cap \mu^{-1}(P_S W^*).$$

Let $\theta \in \mathcal{C}_d^S$. By definition of the S-critical locus,

$$\nabla_\theta \mathcal{L}(\theta) = 0 \qquad \text{and} \qquad \mu(\theta) = P_S W^*.$$

Since $\operatorname{rank}(P_S W^*) = r$, it follows that

$$\operatorname{rank}(\mu(\theta)) = r.$$

It remains to show the cyclic equations with closing arrow $\Sigma_{XY} P_Q$.

For each layer ℓ, the transpose-gradient formula is

$$\big(\nabla_{W_\ell} \mathcal{L}(\theta)\big)^\top = 2 W_{<\ell} \big(\Sigma_X \mu(\theta)^\top - \Sigma_{XY}\big) W_{>\ell}.$$

Using $\mu(\theta) = P_S W^*$, we get

$$\Sigma_X \mu(\theta)^\top - \Sigma_{XY} = \Sigma_X (P_S W^*)^\top - \Sigma_{XY}.$$

Since P_S is an orthogonal projector, $P_S^\top = P_S$, and therefore

$$(P_S W^*)^\top = (W^*)^\top P_S.$$

Moreover, by the definition

$$W^* = \Sigma_{YX}\Sigma_X^{-1},$$

and by symmetry of Σ_X, we have

$$\Sigma_X(W^*)^\top = \Sigma_X(\Sigma_X^{-1})^\top\Sigma_{YX}^\top = \Sigma_X\Sigma_X^{-1}\Sigma_{XY} = \Sigma_{XY}.$$

Hence

$$\Sigma_X(P_SW^*)^\top - \Sigma_{XY} = \Sigma_X(W^*)^\top P_S - \Sigma_{XY} = \Sigma_{XY}P_S - \Sigma_{XY}.$$

Since $P_Q = I - P_S$, this becomes

$$\Sigma_X(P_SW^*)^\top - \Sigma_{XY} = -\Sigma_{XY}P_Q.$$

Substituting into the gradient formula gives

$$\big(\nabla_{W_\ell}\mathcal{L}(\theta)\big)^\top = -2W_{<\ell}\Sigma_{XY}P_QW_{>\ell}.$$

Because $\theta \in \mathcal{C}_d^S$, every partial gradient vanishes. Therefore

$$W_{<\ell}\Sigma_{XY}P_QW_{>\ell} = 0 \qquad \text{for all } 1 \le \ell \le L.$$

Thus

$$(\theta, \Sigma_{XY}P_Q) \in \Gamma_d^r,$$

or equivalently,

$$\theta \in \Gamma_d^S.$$

Since also $\mu(\theta) = P_SW^*$, we have

$$\theta \in \Gamma_d^S \cap \mu^{-1}(P_SW^*).$$

Conversely, suppose that

$$\theta \in \Gamma_d^S \cap \mu^{-1}(P_SW^*).$$

Then

$$\mu(\theta) = P_SW^*$$

and, because $\theta \in \Gamma_d^S$,

$$W_{<\ell}\Sigma_{XY}P_QW_{>\ell} = 0 \qquad \text{for all } 1 \le \ell \le L.$$

As above,

$$\Sigma_X\mu(\theta)^\top - \Sigma_{XY} = -\Sigma_{XY}P_Q.$$

Therefore, for every ℓ,

$$\big(\nabla_{W_\ell}\mathcal{L}(\theta)\big)^\top = -2W_{<\ell}\Sigma_{XY}P_QW_{>\ell} = 0.$$

Hence

$$\nabla_{W_\ell}\mathcal{L}(\theta) = 0 \qquad \text{for every } 1 \le \ell \le L.$$

So

$$\nabla_\theta\mathcal{L}(\theta) = 0.$$

Together with $\mu(\theta) = P_SW^*$, this gives

$$\theta \in \mathcal{C}_d^S.$$

This proves the reverse inclusion and therefore the proposition.

8.2 Detailed proof of Proposition 4.2

Assume that

$$\Gamma_d^r = \bigsqcup_{\rho \in R} O_\rho^r$$

is a finite decomposition into locally closed subsets of Γ_d^r, and that the pieces satisfy

$$\overline{O_\rho^r}^{\Gamma_d^r} \subseteq \bigcup_{\rho' \leq \rho} O_{\rho'}^r.$$

For each $\rho \in R$, define

$$\mathcal{C}_\rho^S := \iota_S^{-1}(O_\rho^r) \cap \mu^{-1}(P_S W^*).$$

We first prove that the pieces $\mathcal{C}_\rho^S$ give a disjoint decomposition of $\mathcal{C}_d^S$. By Proposition 4.1,

$$\mathcal{C}_d^S = \iota_S^{-1}(\Gamma_d^r) \cap \mu^{-1}(P_S W^*).$$

Since

$$\Gamma_d^r = \bigsqcup_{\rho \in R} O_\rho^r,$$

we have

$$\iota_S^{-1}(\Gamma_d^r) = \iota_S^{-1}\left(\bigsqcup_{\rho \in R} O_\rho^r\right).$$

Preimages commute with arbitrary unions, and preimages of disjoint sets are disjoint. Hence

$$\iota_S^{-1}(\Gamma_d^r) = \bigsqcup_{\rho \in R} \iota_S^{-1}(O_\rho^r).$$

Intersecting both sides with $\mu^{-1}(P_S W^*)$ gives

$$\mathcal{C}_d^S = \left(\bigsqcup_{\rho \in R} \iota_S^{-1}(O_\rho^r)\right) \cap \mu^{-1}(P_S W^*).$$

Therefore

$$\mathcal{C}_d^S = \bigsqcup_{\rho \in R} \left(\iota_S^{-1}(O_\rho^r) \cap \mu^{-1}(P_S W^*)\right) = \bigsqcup_{\rho \in R} \mathcal{C}_\rho^S.$$

After discarding the empty terms, this is the claimed decomposition.

Next we prove local closedness. Each O_ρ^r is locally closed in Γ_d^r. Since Γ_d^r is a locally closed subset of $\Omega \times \mathrm{Mat}_{d_0 \times d_L}$, each O_ρ^r is locally closed in $\Omega \times \mathrm{Mat}_{d_0 \times d_L}$. The map

$$\iota_S : \Omega \longrightarrow \Omega \times \mathrm{Mat}_{d_0 \times d_L}$$

is continuous, so

$$\iota_S^{-1}(O_\rho^r)$$

is locally closed in Ω. The map $\mu : \Omega \to \mathrm{Mat}_{d_L \times d_0}$ is regular, hence continuous, and the point $\{P_S W^*\}$ is closed. Therefore

$$\mu^{-1}(P_S W^*)$$

is closed in Ω. Consequently

$$\mathcal{C}_\rho^S = \iota_S^{-1}(O_\rho^r) \cap \mu^{-1}(P_S W^*)$$

is locally closed in Ω. Since

$$\mathcal{C}_\rho^S \subseteq \mathcal{C}_d^S,$$

it is also locally closed in $\mathcal{C}_d^S$ with the subspace topology.

It remains to prove the frontier inclusion

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} \subseteq \bigcup_{\rho' \leq \rho} \mathcal{C}_{\rho'}^S.$$

We use the following elementary fact. If $f : X \to Y$ is continuous and $A \subseteq Y$, then

$$\overline{f^{-1}(A)}^X \subseteq f^{-1}\left(\overline{A}^Y\right).$$

Indeed, $f^{-1}(\overline{A}^Y)$ is closed in X and contains $f^{-1}(A)$, so it contains the closure of $f^{-1}(A)$.

Apply this fact to the continuous map

$$\iota_S|_{\mathcal{C}_d^S} : \mathcal{C}_d^S \longrightarrow \Gamma_d^r.$$

This restriction is well-defined because, by Proposition 4.1,

$$\mathcal{C}_d^S \subseteq \Gamma_d^S = \iota_S^{-1}(\Gamma_d^r),$$

so $\iota_S(\theta) \in \Gamma_d^r$ for every $\theta \in \mathcal{C}_d^S$.

Now observe that

$$\mathcal{C}_\rho^S = (\iota_S|_{\mathcal{C}_d^S})^{-1}(O_\rho^r).$$

Therefore

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} = \overline{(\iota_S|_{\mathcal{C}_d^S})^{-1}(O_\rho^r)}^{\mathcal{C}_d^S} \subseteq (\iota_S|_{\mathcal{C}_d^S})^{-1}\left(\overline{O_\rho^r}^{\Gamma_d^r}\right).$$

Using the assumed frontier inclusion,

$$\overline{O_\rho^r}^{\Gamma_d^r} \subseteq \bigcup_{\rho' \leq \rho} O_{\rho'}^r,$$

we obtain

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} \subseteq (\iota_S|_{\mathcal{C}_d^S})^{-1}\left(\bigcup_{\rho' \leq \rho} O_{\rho'}^r\right).$$

Preimages commute with unions, so

$$(\iota_S|_{\mathcal{C}_d^S})^{-1}\left(\bigcup_{\rho' \leq \rho} O_{\rho'}^r\right) = \bigcup_{\rho' \leq \rho} (\iota_S|_{\mathcal{C}_d^S})^{-1}(O_{\rho'}^r).$$

But

$$(\iota_S|_{\mathcal{C}_d^S})^{-1}(O_{\rho'}^r) = \mathcal{C}_{\rho'}^S.$$

Hence

$$\overline{\mathcal{C}_\rho^S}^{\mathcal{C}_d^S} \subseteq \bigcup_{\rho' \leq \rho} \mathcal{C}_{\rho'}^S.$$

This proves the proposition.

The argument gives an inclusion rather than an equality. The possible failure of equality comes from the strictness of

$$\overline{f^{-1}(A)}^{X} \subseteq f^{-1}\left(\overline{A}^{Y}\right)$$

for a general continuous map f. In the present setting, a point in $\iota_S^{-1}(\overline{O_\rho^r}^{\Gamma_d^r})$ may be reachable as a degeneration of O_ρ^r in the full cyclic variety while not being reachable by a degeneration that remains inside the fixed specialization determined by ι_S. Therefore the natural statement for the S-critical locus is the one-sided frontier condition above.

9 Deprecated